Pergamon

S0360-1315(96)00007-3

Computers Educ. Vol. 26, No. 1-3, pp. 131-134, 1996

0360-1315/96 $15.00 + 0.00

MANAGING TECHNOLOGICAL CHANGE IN EDUCATION—WHAT LESSONS CAN WE ALL LEARN?

RACHAEL SCOTT and BRENT ROBINSON

Department of Education, University of Cambridge, 17 Trumpington Street, Cambridge CB2 1QA, England
[*Fax*: 01223 332894; *e-mail*: RS208@.CAM.AC.UK]

Abstract—Technological innovation within higher education is currently proceeding at an unprecedented pace, stimulated by a desire to improve the quality of teaching and learning and to generate citizens, workers and life long learners with the skills to maximize the potential which will be afforded by information technology (IT). But it is not in higher education alone that such technological innovation is taking place. Within the school sector there has been extensive and systemic technological change over a 15 year period. This paper explores some of the lessons which have been learned about effective technological innovation, notably from the extensive application of information technology in the schools sector, which appear to have importance for the implementation of IT supported teaching and learning in higher education. Copyright © 1996 Elsevier Science Ltd

Early IT initiatives in U.K. schools were essentially technology centred. Research on change of this kind in schools in the 1970s and early 80s focused on technical aspects on innovations [1]. A majority of the studies of that period concentrated on the resistance of individual teachers, which was perceived as the main factor in the process of change. Overcoming this resistance was seen as dependent on providing technological access and showing the technological advantages of the proposed innovation. The principle seemed to be that "as long as the facilities are available and teachers are trained in computing, adoption . . . is inevitable" [2, p. 229]. Initially, even the facilities were scarce. Only a single computer was issued to each school with a very limited range of software. Later, the Government funded more hardware for schools and the ratio of computers to students improved. Similarly, the Government funded the development of new educational software. The Microelectronics Education Programme and the Microelectronics Education Support Unit were both responsible for the development of a wide range of software for use across the subjects and age phases of compulsory education. But it became evident that access to hardware and software alone are insufficient. Learning how to use information technology in the classroom involves more than training in hardware and software use. It requires pedagogic understanding of what CAL applications are trying to do and of what the hardware and software are capable of doing. There has to be a match between the teaching and learning style of the classroom and what the computer has been programmed to do or what, with more generic applications, the technology is to be used for. Herein, there lies a challenge. While the computer is a protean machine, capable of being deployed to match a variety of teaching and learning styles and to fulfil a range of educational goals, it also has potential for enlarging and changing educational processes and aims. The use of IT in education therefore involves not just a change of teaching resources but also of teaching strategies and beliefs. Fullan [3] has identified the fact that all three of these are major dimensions of change and he warns that change must occur in all dimensions if educational innovation is to be successful:

> "Implementation involves the development of new teaching approaches and examination of underlying beliefs . . . most [change] efforts . . . have concentrated on 'paper' changes . . . [overlooking] people (behaviour, beliefs, skills) in favour of things (regulations, materials) and this is essentially why it fails more times than not . . . people are much more difficult to deal with than things [but] also more necessary for success" (p. 249).

Increasingly, those involved in technological innovation in schools have discovered that human factors are important variables in the change process. The advent of the National Curriculum in 1988 forced all schools to consider how best to manage the introduction of information technology and increasing attention has been necessary to human and institutional as well as to technological issues. The Education Reform Act of 1988 mandated schools to use information technology in the delivery of the curriculum: "Our aim is to ensure that all pupils have a firm foundation of general information technology skills which can be developed through their work in subjects across the curriculum..." [4, para 3.9]. In order for this to be achieved, not only individual teacher expertise but whole school co-ordination was required and schools were urged: "... to ensure that a coherent and consistent policy for information technology exists" [5, para. 3.3]. In the last 6 years schools themselves and educational support agencies (notably the local education authorities and the National Council for Educational Technology) have increasingly turned to the personal and institutional factors involved in technological change. For examples of the literature and support materials produced see NCET [6–8]; the STAC Project [9,10]; the University of Sussex/ Brighton Polytechnic Whole School IT Development Project [11] and the Managing IT in Schools Project [12].

But it was not only schools as institutions which had to draw up and enact these changes. New entrants to the school teaching profession were to be trained to use IT too so the mandate extended also to higher education institutions which carried out teacher training. Higher education training courses had to contain: "... compulsory and clearly identifiable elements which enable students to make effective use of information technology in the classroom and provide a sound basis for their subsequent development in this field" [13, para 6.6]. As a result, the National Council for Educational Technology has been active through its funding of Project Intent [14] to explore how a technologically rich environment might be initiated and managed among academics in teacher training institutions.

Concurrent with the shift away from a technocentric view of technological innovation, was the growth internationally of a substantial body of literature concerning the nature of educational change in schools [15] and technological innovation in schools came to be seen as part of the broader field of educational change. According to Cox and Rhodes [16]: "It has been recognised that many of the barriers to ... the adoption of microcomputers are specific examples of the barriers to ... change in general". This suggests that a broad approach to the study of issues involved in using computers in education is warranted.

While technological innovation in schools is now adopting this wider perspective, there appears to be little evidence that the higher education community is doing likewise.

Indeed, recognition of the application of change theory to the introduction of technology within tertiary education is not ubiquitous. Rather, such research reflects more isolated cases than generalizable knowledge [14, 17, 18]. Thus, in an influential report to the Scottish University Principals, McFarlane [19] rightly points out that whilst the implementation of such a technological programme of teaching would require the application of well known procedures associated with the management of change, as yet no such theory exists for higher education.

The very fact that this dearth has been identified will probably mean that such a theory will evolve, as those in the field realize the need and potential value of such work. However, this important lacuna can be partly remedied by looking to the substantial experience and expertise of school-based technological change and further to the lessons from the past that the higher education milieu has already generated.

Comparable to educational technology's school level histography, in the last 20 years there have been a succession of government funded initiatives to investigate and develop computer-based teaching and learning at university level. This began in 1974 with the National Development Programme for Computer Assisted Learning and up to the present date has included such projects as the Computers in Teaching Initiative (CTI), The Information Technology Training Initiative (ITTI), The Joint Academic Network (JANET), and the Teaching and Learning Technology Programme (TLTP).

Higher education has therefore benefited from these projects and the substantial funding ploughed into this area. Yet whilst these initiatives had certainly been instrumental in building up necessary knowledge and new technological developments, they have not resulted in the prolific or in some instances even apparent utilization of information technology within the wider context of higher education.

A greater conceptual and procedural knowledge of the change process helps in moving towards an understanding of this situation. As has been stipulated earlier in this paper, for educational innovation to be successful, change has to take place in materials, teaching approaches and beliefs [3]. Thus, with regard to technological change at the school level, the three successive phases of concentration on technology, pedagogy and "people" can respectively be said to exemplify the multidimensional nature of change.

However, it is our contention that higher education is really only just beginning to address the second of these "elements" and significantly has not considered the third. According to this line of thought, it is because of this negligence that technology has not always fully realized its projected ideal.

Historically, those in the field have been far more concerned with either developing and exclaiming the merits of new and better technology or of detailing reasons why educational technology is not fulfilling its "promise" [20–30]. Instead we should be learning from those currently leading in the field, the schools, and also concentrating on the other two of the three composite elements of the change process.

Definitely, there has been a more promising move towards the teaching and learning potential that information technology can provide, with for example, one of the governments more recent projects, the TLTP, being aimed at developing courseware for flexible and open teaching using IT. It is perhaps significant to here once again recap on the central tenet of change theory. For Fullan, teaching strategies actually have to change and this should be the predominant way in which technology is employed. Thus, a notion of "exploitation" (of the technology) as opposed to "accommodation" should prevail [19, 31]. Currently, however, this "Educational Renaissance" is not apparent. Rather, teaching and learning within higher education remains in an outdated seventeenth century mode, despite the advent of computers and particularly multimedia potential. The norm for computing across campuses, the 24 hr computer lab, primarily used for word processing purposes shows no mark of progression for education. The central message we should therefore be gleaning from the schools and adopting for our own purposes within higher education is a more concerted effort and concentration on learning as opposed to the technology itself.

Significantly, this notion fits comfortably with the government's proposal to turn Britain into a "Learning Society" by the year 2000. Universities as "learning institutions' will have to turn their attention to something that has previously only received lip service—i.e. learning itself [25].

Perhaps the most important lesson to be learnt from the experience of school-based technological change is the move towards a recognition of the importance of "people". Significantly, within organizational and business literature this has also been the case [32]. It is important to realize that it is the meanings, the understandings to which individuals attach regarding the change that are paramount. For, real change can only occur if the individuals' perceptions alter [3, 33]. Further, the "power" an individual can possess as a change agent [33] needs to be recognized as a feature of significant leverage within an institutional change effort.

Comparable themes of user involvement in order to develop a sense of "ownership", relating the innovation to needs and concerns and other tried and tested change strategies, although not new, are of importance and need to be re-examined in light of this more phenomenological approach.

In short, an understanding of the phenomenology of technological innovation in particular, and of educational change in general would mean that an individual would possess the capacity to know when and how to pursue and implement possible innovation and significantly, know when to reject others [3].

The change perspective centres upon an awareness of the change process along with adherence to the key factors known to impact on the probability of success. As Fullan [3, p. 93] contends

> "The 'solution' to the management of educational change is straightforward. All we need to do in any situation is to take the factors and themes described ... change them in a positive direction, and then orchestrate them so that they work smoothly together .... To bring about more effective change, we need to be able to explain not only what causes it but how to influence those causes".

Here, Fullan is fundamentally distinguishing between what changes to implement (theories of education) and how to implement these changes (theories of change). This distinction is illuminated to a greater degree by comparing and contrasting a theory of change (an awareness of "what" and "why") with a theory of changing (an awareness of "how") [3, 27, 33].

However, both at the primary, secondary and tertiary levels of education there has been an overt tendency to concentrate on the former as opposed to the latter. Yet the knowledge base necessary to build upon towards reaching a theory of changing, at both levels of education, is already in place. It is time to move beyond the technocentric place of IT innovation in higher education. Our collective experience of the past (and present) should be re-examined in the context of a change paradigm in order to inform and generate pedagogical and "people"; oriented strategies for the future—a theory of changing.

REFERENCES

1. Grunberg J. and Summers M., Computer innovation in schools: a review of selected research literature. *J. Inform. Technol. Teacher Educ.* **1,** 255–276 (1992).
2. Anderson R., Hansen T., Johnson D. and Klassen D., Instructional computing: acceptance and rejection by secondary school teachers. *Sociol. Work Occup.* **6,** 227–250 (1979).
3. Fullan M., *The New Meaning of Educational Change.* Teachers College Press, New York (1991).
4. DES, *Design and Technology for Ages 5 to 16.* Department of Education and Science, London (1989).
5. DES, *Design and Technology Working Group Interim Report.* Department of Education and Science, London (1988).
6. National Council for Educational Technology, *Supporting Teacher Change.* NCET, Coventry (1989).
7. National Council for Educational Technology, *Working with Teachers.* NCET, Coventry (1989).
8. National Council for Educational Technology, *Managing IT.* NCET, Coventry (1995).
9. Passey D. and Ridgway J., *Co-Ordinating National Curriculum IT: Strategies for Whole School Development.* Framework Press, Lancaster (1992).
10. Passey D. and Ridgway J., *Effective In-Service Education for Teachers.* NORICC, Newcastle-upon-Tyne (1992).
11. Erault M. *et al., Whole School IT Development.* IT in Schools County Support Team, Sittingbourne, Kent (1991).
12. North R., *Managing IT.* University of Ulster, Londonderry (1990).
13. DES, *Circular 24/89 Initial Teacher Training: Approval of Courses.* Department of Education and Science, London (1989).
14. Somekh B., A research approach to IT development in initial teacher education. *J. Inform. Technol. Teacher Educ.* **1,** 83–99 (1992).
15. Wu P., Why is change difficult? Lessons for staff development. *J. Staff Devel.* **9,** No. 2, 10–14 (1988).
16. Cox M. and Rhodes V., The uptake and usage of microcomputers in primary schools with special reference to teacher training. ESRC Research Report, InTER/8/89, Lancaster (1989).
17. Rutherford D., Appraisal in action: a case study of innovation and leadership. *Stud. Higher Educ.* **17,** 201–445 (1992).
18. Lieblum M. D., Implementing CAL at a university. *Computers Educ.* **18,** 109–118 (1992).
19. McFarlane A., Teaching an learning in an expanding higher education system. The Committee of Scottish University Principals (1992).
20. Dickerson M. D. and Gentry J. W., Characteristics of adopters and non-adopters of home computers. *J. Consumer Res.* **10,** 225–235 (1983).
21. Butler D. L., Interests in and barriers to using computers in instruction. *Teach. Psychol.* **13,** 20–23 (1986).
22. Faseyitan S. O. and Hirschbuhl J., Computers in university instruction: what are the significant variables that influence adoption? *Interactive Learn. Int.* **8,** 185–194 (1992).
23. Gardner N., Integrating computers into the university curriculum: the experience of the U.K. computers in teaching initiative. *Computers in Educ.* **12,** 23–27 (1988).
24. Gressard C. and Loyd B. H., Age and staff experience with computers as factors affecting teacher attitudes toward computers. *School Sci. Math.* **85,** 203–209 (1985).
25. Hammond N., Gardner N., Heath S., Kibby M., Mayes T., McAleese R., Mullings C. and Trapp A., Blocks to the effective use of information technology in higher education. *Computers Educ.* **18,** 155–162 (1992).
26. Hill T., Smith N. D. and Mann M. F., Role of efficacy expectations in predicting the decision to use advanced technologies: the case of computers. *J. appl. Psychol.* **72,** 307–313 (1987).
27. Jones P. and Lewis J., Implementing a strategy for collective change in higher education. *Stud. Higher Educ.* **16,** 51–61 (1991).
28. Mudd S. and McGrath K., Correlates of the adoption of curriculum integrated computing in higher education. *Computers Educ.* **12,** 457–463 (1988).
29. Nelson R. B. and Erwin T. D., Examining the organizational characteristics of CAI projects in instructions of higher education. *J. Educ. Technol. Syst.* **14,** 297–305 (1985).
30. Spuck D. W., An analysis of the cost-effectiveness of CAI and factors associated with its successful implementation in higher education. *AEDS J.* **6,** No. 4, 4–10 (1981).
31. Laurillard D., *Rethinking University Teaching—A Framework for the Effective Use of Educational Technology.* Routledge, New York (1993).
32. Kanter R. M., *The Change Masters.* Routledge, New York (1984).
33. Fung A. C. W., Management of educational innovations. *The 7th Regional Conference of Commonwealth Council for Educational Administration.* Hong Kong (1992).